

AI LITERACY

FOR STUDENTS

Understanding How AI Works — and How to Use It Well

Leadership By Design Certification Program

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WHERE WARRIORS BELONG · EAST STROUDSBURG UNIVERSITY

WHAT WE'LL COVER TODAY

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How AI Thinks

Tokens, context windows, training data & hallucination

02

How You Interact

Prompts, iteration, multimodal & guardrails

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How It's Organized

LLMs, models, APIs, memory & agents

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Platform Comparison

ChatGPT vs Claude vs Gemini — what matters

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Ethical Considerations

Privacy, bias, transparency & responsible use

WHAT IS GENERATIVE AI?

GENERATIVE AI

AI that creates new content — text, images, audio, video — based on patterns learned from training data.

Unlike traditional software that follows fixed rules, generative AI produces original outputs in response to your prompts.

EXAMPLES YOU MAY ALREADY KNOW

ChatGPT

OpenAI — text, images, audio

Claude

Anthropic — text, documents

Gemini

Google — text, images, video

Copilot

Microsoft — integrated in Office

Grok

xAI — built into X / Twitter

TOKENS & CONTEXT WINDOW

THINK OF IT LIKE THIS:

A token is a small chunk of text. Context window = total working memory (input + output combined). Max input = what you can send in. Max output = how much the AI can write back.

Platform	Context Window	Max Input	Max Output	Key Takeaway
ChatGPT (GPT-5)	~128K	~128K	~128K	Standard context, large output
Claude (Sonnet 4.6)	~200K	~200K	~64K	Large context, strong output
Gemini 2.5 Pro	~1M	~1M	~65K	Best for ingesting huge docs
Copilot	~128K	~128K	~16K	Standard Office integration
Grok	~128K	~128K	~16K	Standard
DeepSeek	~128K	~128K	~16K	Open-source, lower cost

HOW AI THINKS: KEY CONCEPTS

TRAINING DATA

The massive collection of text AI learned from before you ever talked to it — books, websites, articles. This is what shaped how it responds.

HALLUCINATION

When AI confidently states something completely made up. It's not lying — it fills gaps in ways that sound plausible but aren't accurate. Always verify important facts.

TEMPERATURE

Controls creativity vs. precision. High temperature = more creative and unpredictable. Low temperature = more focused and consistent.

KNOWLEDGE CUTOFF

The date AI stopped learning new information. ChatGPT: Early 2025. Claude: Aug 2025. Gemini: Early 2025. Copilot & Grok use live data.

PROMPTING & HOW YOU INTERACT

PROMPT — The instruction or question you give the AI. Better prompts get better results. Think of it like giving directions — the more specific you are, the closer you get to your destination.

1 START

Ask your question or give your instruction. Be specific about format, audience, and purpose.

2 EVALUATE

Read the response critically. Does it do what you need? What's missing or off?

3 ITERATE

Refine your prompt. Ask it to adjust tone, length, add examples, or try a different approach.

NLP (Natural Language Processing) allows AI to understand human language — why you can talk to it just like a person. Multimodal = AI that handles more than text (images, audio, video).

LLM

Large Language Model

The technical name for AI like ChatGPT, Claude, and Gemini — trained on enormous amounts of language data.

Model / Version

Different releases of the same AI

ChatGPT has GPT-5 (its current flagship). Claude has Sonnet 4.6. Gemini has 2.5 Pro. Each version usually improves on the last.

API

Application Programming Interface

A 'back door' that lets developers connect AI into their own apps. A university could use the ChatGPT API to power an automated tutoring tool without students visiting ChatGPT.com.

System Prompt

Hidden setup instructions

Instructions given to the AI before a conversation starts, used to customize behavior — like telling AI to always respond in an academic tone.

PROJECTS, ARTIFACTS, CANVAS & MEMORY

PROJECTS

Organize conversations around a specific goal with shared context.

Claude: ✓ Yes
ChatGPT: ✓ Yes
Gemini: Gems
Copilot: ✗ No
Grok: ✗ No

ARTIFACTS vs. CANVAS

Similar idea, different experience.

Claude Artifacts: A rendered output (doc, code, HTML) in a side panel — tied to the conversation, not a standalone file.
ChatGPT Canvas: A live co-editing space — you and the AI edit the same document together in real time.
Key difference: Artifacts = you prompt to change it. Canvas = you can edit it directly.
Gemini: No equivalent yet. Copilot: Built into Word & Excel.

MEMORY

AI retains info from past conversations to personalize future ones.

ChatGPT: ✓ Viewable & editable
Claude: ✓ Limited user control
Gemini: ✓ Tied to Google account
Copilot: Limited
DeepSeek: ✗ No

 **ETHICS:** Memory raises questions about privacy and transparency. Users should know what's stored and have control over it.

AGENTS, FINE-TUNING & RAG

AGENTS

AI that takes actions on your behalf — not just answers questions. It can click, search, fill forms, and execute multi-step tasks.

ChatGPT: Operator / GPT Actions

Claude: Computer Use (beta)

Gemini: Advanced Agents

Copilot: Copilot Agents (M365)

FINE-TUNING

Training an AI further on specific data to make it better at a particular domain.

Example: A hospital might fine-tune AI on medical records to improve responses for doctors.

RAG

Retrieval-Augmented Generation — AI pulls in real-time or specific documents before responding, rather than relying only on training data.

Like giving the AI a reference book to consult while it answers your question.

PLATFORM COMPARISON AT A GLANCE

Feature	ChatGPT	Claude	Gemini	Copilot	Grok
Parent Co.	OpenAI (Microsoft)	Anthropic	Google	Microsoft	xAI (Musk)
Best Model	GPT-5	Sonnet 4.6	2.5 Pro	GPT-5	Grok 4
Context Window	~128K	~200K	~1M	~128K	~128K
Knowledge Cutoff	Early 2025	Aug 2025	Early 2025	Live web	Live X data
Image Input	✓	✓	✓	✓	✓
Voice	✓	✓ (new)	✓	✓	X
Projects	✓	✓	Gems	X	X
Artifacts/Canvas	Canvas	Artifacts	X	Office	X
Memory	✓ Full	✓ Limited	✓ Google	Limited	X

ETHICAL CONSIDERATIONS

HALLUCINATION & ACCURACY

AI can sound confident and still be wrong. Always verify claims against primary sources before submitting any work.

PRIVACY & DATA

What you type into AI may be stored and used for training. Don't enter personal information, passwords, or anything you wouldn't post publicly.

BIAS IN TRAINING DATA

AI reflects the biases in its training data. Outputs may skew toward dominant perspectives. Think critically — don't take AI responses at face value.

ACADEMIC INTEGRITY

Know your course policies on AI use — they vary by professor and assignment. When in doubt, ask. Using AI without disclosure where it's not permitted is a violation of academic integrity.

TRANSPARENCY

Be transparent when AI shapes your work. Engage with it critically — use it to think better, not to avoid thinking. Your voice and judgment still matter most.

AI IN YOUR STUDIES — PRACTICAL IDEAS

STUDY & NOTE-TAKING

Use AI to summarize lecture notes, explain confusing concepts in plain language, or quiz yourself before an exam. You still need to understand the material — AI helps you get there.

WRITING & FEEDBACK

Ask AI to give feedback on a draft, suggest stronger arguments, or flag unclear sentences. Treat it like a writing tutor — not a ghostwriter. Your ideas should still drive the work.

RESEARCH SUPPORT

AI can help you find angles on a research topic, understand dense academic articles, or build an outline before you start writing. Always verify sources — AI can hallucinate citations.

CAREER & SKILL BUILDING

Use AI to practice interview questions, polish your resume, draft cover letters, or learn a new skill. AI fluency is itself a career advantage — knowing how to prompt well sets you apart.

ACCESSIBILITY

AI can help you work through content in different ways — simplify dense readings, translate concepts into your first language, or provide step-by-step explanations when you're stuck.

KNOW THE RULES

AI policies vary by professor, course, and assignment. Read your syllabus, ask questions early, and document how you used AI when required. When in doubt, disclose.

QUESTIONS & DISCUSSION

Next: Prompting Strategies for Learning & Academic Success

TRY IT

[claude.ai](#) · [chatgpt.com](#) · [gemini.google.com](#)

CONTACT

Your AI Literacy Workshop Facilitator